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RESEARCH ARTICLE

Data-driven placemaking: Public space canopy design through multi-objective optimisation considering shading, structural and social performance

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Abstract In the context of ongoing densification of cities and aging urban populations, public spaces are a crucial infrastructure to support the physical and mental wellbeing of urban residents. The design of public space furniture elements is often standardised, and not considered in relation to environmental conditions and mechanisms of social interaction. This article presents a digital workflow to generate site-specific designs for shaded public seating, considering the relationships of local public places to their surroundings. A strategy for customised and site-specific design is developed through the use of multiple software tools, employing evolutionary algorithms and multi-objective optimisation. The method is applied to a small public space canopy prototype installed within a public housing estate in Hong Kong, incorporating additional criteria to achieve a low-cost and light-weight structure. Through multiple stages of refinement and optimisation, a material, structural and social performance-driven outcome was achieved that creates a shaded space for public seating, people watching and social interaction. As part of a larger research agenda exploring architectural form-finding and environmental psychology, the project represents potential new applications in the emerging field of socially driven computational design.

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1. Introduction

As cities are becoming increasingly dense to accommodate population growth and limit their impact on surrounding ecosystems, the design of urban public spaces is of crucial importance to the quality of urban environments, experiences and the social activities of everyday life. Private living spaces become less affordable and spacious, and shared spaces around residential buildings are used as part of important socialising, recreational and community activities. Studies have shown that the quality of life of vulnerable social groups such as low-income families or elderly can be significantly improved through good quality public facilities (Saunders et al., 2014; Gou et al., 2018). In high-density cities such as Hong Kong, public open spaces with good environmental comfort and opportunities for social interaction can contribute to active aging, social integration, and to the physical and psychological health of elderly residents (Yung, 2016; Lau and Murie, 2017; Zheng et al., 2015).

The layout of urban space affects social dynamics, as it "facilitates co-presence and regulates interpersonal relationships" (Madanipour, 2003, p. 206). People are more inclined to inhabit or socialise in urban spaces that have good visibility towards their surroundings (Gehl, 1987; Whyte, 1980), and that offer privacy and protection (Appleton, 1975, 1984). Thus, the manipulation of spatial relationships in urban spaces through landscape or architectural elements can help create conditions conducive to socialising (Batty, 2001; Stamps, 2005). Research has shown that morphological aspects such as the 'edge effect' (Gehl, 2010), and the concept of 'triangulation' - where urban elements serve as a conversation starter, can stimulate social interaction between strangers (Whyte, 1980). Chance encounters and casual neighbouring can lead to social integration, attachment to place, and the formation of supportive communities (Low, 2000; Talen, 1999).

People's willingness to inhabit public spaces is also strongly affected by environmental conditions, especially in hot summer conditions in cities such as Hong Kong (Ng and Cheng, 2012). Research has determined that shading and wind can improve human thermal comfort (Li, et al., 2018) and that landscape elements that improve shading and wind flow through urban public spaces can lead to increased social activities (Huang, et al., 2016). However, the design of public space furniture elements is often standardised, and their location, orientation and formal qualities are often not considered in relation to the environmental conditions and social interaction of the sites in which they are placed (Lee et al., 2013; Mesthrige and Cheung, 2020).

This article focuses on site-specific shaded public seating, designed in consideration of the relationships of specific places to their surroundings. It presents a novel methodology for generating customised public space canopies, combining architectural form-finding and structural optimisation with a data-driven approach to analysing urban visibility and environmental conditions. The study involved quantifying qualitative aspects of how people experience public space, comfort and social relations, and thus aimed to integrate some of the technical and psychological design aspects of small urban spaces into a comprehensive computational process. A customised methodology and workflow were developed through the integration of multiple software tools, exploring the use of evolutionary algorithms and multi-objective optimisation to produce a cost effective, structurally, socially, and environmentally performative solution.

1.1. Form-finding and performative design in architecture

Architectural form-finding is a broad concept that has most often been associated with the self-optimisation of material geometries and structures in relation to structural performance (Adriaenssens et al., 2014; Turrin et al., 2011). The term has been associated with lightweight architecture and tensile membrane structures developed by Frei Otto and others (Meissner and Möller, 2015) and the catenary structures of Gaudi (Huerta, 2006), Isler (Chilton and Isler. 2000) or recently the Block Research Group (Popescu et al., 2021). However, in recent year the notion a single, optimised outcome at the equilibrium between geometrical, structural and material properties (Burkhardt, 2016) has been expanded to include multiple objectives and a more complex solution space.

Form-finding processes have since long been performed through computational tools using physics-based algorithms (Ahlquist and Menges, 2011), which enable the combination of the previously separate processes of form generation and evaluation (Nguyen et al., 2020) and transform form-finding into an interactive dynamic process (Attar et al., 2010). The use of the term 'form-finding' has been expanded to include other criteria, such as the cost optimisation of structures (Ekici et al., 2015), wind performance (Yao et al., 2018), construction and shading (Agirbas, 2019). These approaches expand the notion of form-finding towards 'performative design optimisation', which has been defined by Oxman (2008) as an approach which integrates a generative design process within a simulated environment in order to evaluate the design's performative capabilities in relation to that environment.

In this paper, we present the novel concept of form-finding as a generative process that considers an architectural membrane structure in relation to its environment's specific social interaction and sun shading requirements. The work connects existing form-finding methodologies to urban environment and visibility analysis, combining an integrated computational modelling and simulation environment with algorithms for multi-objective optimisation.

In recent years, research around computational performance optimisation has increased significantly to include a range of optimisation workflows and performance evaluation criteria, using various swarm and evolutionary optimisation algorithms (Ekici et al., 2019). These algorithms can address complex architectural design challenges, using the process of Multi-Objective Optimisation (MOO) to explore Multiple-criteria decision-making (MCDM) problems which contain two or more optimisation criteria that are conflicting: if a solution is improved towards one of the goals, other criteria are reduced (Harkouss et al., 2018). Instead of a single solution, there are a number of "Pareto optimal" solutions (Miettinen, 1999). The analysis of the solution space between these objectives helps to identify and quantify the trade-offs between and demonstrate the consequences of prioritising each of the different criteria (Navarro-Mateu et al., 2018).

Examples of the wide range of architectural applications are exploring the trade-offs between architecture functionality, structural and environmental performance of a stadium design (Yang et al., 2018), energy and daylight optimisation of façade shading devices (Kirimtat, 2019), and the testing of residential neighbourhood layouts in relation to wind and microclimate comfort (Wu et al., 2021). Scholars have defined the combined of MOO with the paradigm of performance simulation evaluation as defined by Oxman (2008), as Simulation-Based Multi-Objective Optimisation (SBMOO) (Nguyen et al., 2014). The study presented here builds on these examples and methods by employing a methodology that evaluates the spatial relationships that are produced by the architectural form, defined in relation to its spatial and social environment. The term 'performance' is expanded with the notion of 'social performance' which is quantified through measuring visibility across urban spaces.

1.2. Placemaking methods in urban design

Since the 1970s, the term 'placemaking' has been increasingly used amongst the social sciences, including within the field of urban planning (Friedmann, 2010). While there is a range of different definitions of the term, in the context of this article we refer to notion of placemaking as the process of designing urban spaces to promote public life and pedestrian activity (Walljasper, 2004). The notion refers to 'place' as opposed to 'space', indicating human attachment to places that are attractive and suitable for their desired modes of use (Schneekloth and Shibley, 1995). Cresswell (2014, p.39) defines 'place' as "constituted through reiterative social practice", emphasizing that the value of a place lies in its ability to stimulate events and social behaviours.

Quantitative research into the relationships between human activities and urban morphology has been conducted since the 1980s (Gehl, 1987; Hillier and Hanson, 1984; Whyte, 1980). A range of methods has been developed around the theories of Space Syntax, incorporating analysis of circulation network configuration and visibility mapping across complex spaces (Batty, 2001; Hillier, 1996). Neurophysiological research and behavioural theory suggest that boundary and visibility conditions regulate social behaviour in urban environments (Stamps, 2005). An increasing number of data-driven studies in recent years has focused both on the evaluation of predictive computational models (Askarizad and Safari, 2020; Karimi, 2018; Loit, 2021) and on empirical studies around the sociability of spaces (Koohsari et al., 2015; Vroman, 2017), confirming the association between visibility and sociability (De Stefani and Mondada, 2018; Zakariya et al., 2014) used as a performance measure in our study. While these insight, relational principles and methodologies are starting to be implemented in generative architectural design (Goldstein et al., 2020), their application towards the design of small-scale urban spaces is rare. This study explores this knowledge gap as part of a wider research agenda exploring environmental psychology and data-driven design strategies for sociable urban spaces.

Related to the sociability of public spaces is the research into urban micro-climate and human comfort. Many studies have been conducted around the relationships between urban morphology and urban ventilation, establishing that urban porosity and greening can increase comfort and pollution dispersion (Abdollahzadeh and Biloria, 2021; Ng, 2009; Ng et al., 2011; Xue et al., 2017). While these studies provide valuable insights around configurational aspects of urban spaces at the larger scale, there is limited research on the formal aspects of urban places at the micro-scale. A number of studies around public seating highlights the importance of wind blocking, people watching and human comfort factors including temperature, solar radiation, and relative humidity (Zhou et al., 2015). Legge (2020) poses that seating design features should accommodate psychological comfort, physical comfort, and pleasure, and Hadavi et al. (2015) found that seating areas which encourage socialising are most preferred. Carroll et al. (2020) studied seating needs of elderly in residential areas and concluded that more sheltered outdoor spaces are needed to avoid seasonal limitations to neighbourhood socialising. Our study incorporates insights from previous studies into its methodology, using the environmental comfort and socialising properties of public seating as objectives for the performance of a generative public seating canopy design.

2. Schematic design methodology

This section describes the case study project used to develop a computational workflow, focusing on an urban site analysis, and the setting up and testing of a multi-objective optimisation process that incorporated site relationship criteria as well as internal performance objectives.

2.1. Case study site analysis and selection

The location for our prototype is a rooftop public space in between a retail centre and several high-rise residential buildings of the Yau Oi Estate in Tuen Mun, Hong Kong, a public housing estate opened in 1982 that accommodates a large proportion of elderly residents. The area was analysed using a series of algorithms including DeCodingSpaces (Abdulmawla et al., 2017), Ladybug (Roudsari, 2013), and customised scripts developed by the authors. The specific site was chosen based on surveillance ('being seen'), visibility ('seeing'), enclosedness, summer sun and connectivity parameters (Fig. 1). Five possible locations were chosen for quantitative analysis and comparison. The locations are indicated in Fig. 1 and their spatial and environmental analysis parameters are given in Table 1. All values are normalised to a 0—1 range except Solar Radiation.

The location for the prototype (Site 4) was chosen due to its good people watching opportunities and as it could benefit the most from blocking views from surrounding buildings to reduce the sensation of 'being seen'. Table 1 shows that Site 4 has high levels of surveillance, medium-high values for visibility and low values for enclosedness, which indicate that the location has good opportunities to survey all of its surroundings. The medium-low levels of

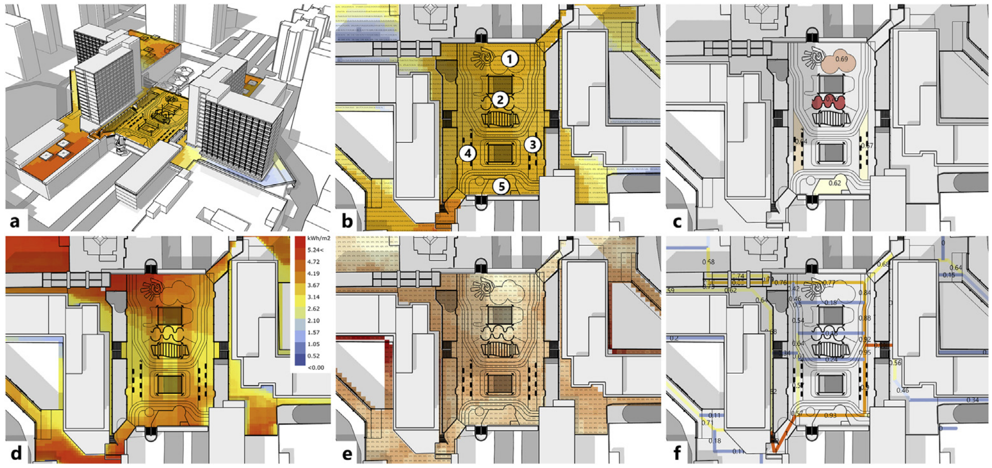


Fig. 1 Public space analysis mapping the levels of (a+b) surveillance, (c) enclosedness, (d) environmental comfort, (e) visibility and (f) connectivity across the different site areas.

connectivity (network integration) show that the location experiences less pedestrian traffic, which makes it attractive to longer periods of resting and socialising. The lower solar radiation value indicates a shorter duration of sunlight exposure, which implies that a smaller size canopy can make a significant impact on the usability of the location as the roof covering can be adjusted to a smaller range of different sunlight vectors.

2.2. Multi-objective optimisation for urban placemaking

As part of the ambition to develop a low-cost and demountable public space canopy system, a tensile membrane with steel support system was chosen for the project. While traditional tensile membrane systems often use poles and cables around the outside of the canopy, a centralised frame principle was chosen to avoid obstacles in the public space and simplify installation on site. While this basic typology was predetermined in relation to budgetary, regulatory and site management constraints, the specific morphology of the canopy on site was subject to a wide range of possibilities.

For our computational workflow, the traditional process of membrane form finding, which has a single optimisation criterion, was extended with additional fitness criteria related to the urban setting such as shading and visibility. Additional project goals such as minimising costs and structural weight meant that a multi-objective optimisation process was necessary. As part of the generative process

Table 1 Site options and measurements, with the selected option in bold.

	Site 1	Site 2	Site 3	Site 4	Site 5
Surveillance	0.15	0.25	0.40	**0.45**	0.30
Enclosedness	0.69	0.76	0.67	**0.64**	0.62
Solar Radiation (kWh/m^2)	4.35	3.14	2.85	**2.80**	3.67
Visibility	0.79	0.75	0.73	**0.76**	0.81
Connectivity	0.77	0.16	0.90	**0.57**	0.93

and multi-criteria decision analysis workflow, Evolutionary Algorithms (EA) were used. The modelling setup was constructed in Rhinoceros 3D with Grasshopper, and using the Galapagos (Rutten, 2013) and Wallacei (Makki et al., 2018) evolutionary solver plugins. The plugins were selected from a range of available optimisation tools available for the Rhino3D/Grasshopper platform, including Octopus (Vierlinger, 2012), SilverEye (Cichocka et al., 2017), and Opossum (Wortmann, 2017) and Optimus (Cubukcuoglu et al., 2019). Galapagos was used in the process design stage due to its user-friendly interface and ability to quickly visualise the flexibility in the generative model setup, while Wallacei was used in the design research and development stage due to its detailed analytical functionalities, and its comprehensive tools for the selection and visualisation of the solution space.

The evolutionary algorithms use several interconnected steps, including defining a parametric definition of a genotype (the base typology described above), the generation of a population of individuals (phenotypes) containing random variations and mutations, and the evaluation of these using fitness objectives. The algorithms select the top performing individuals to generate a new set of improved individuals, across a predefined number of successive generations. The Galapagos software uses Simulated Evolution and Simulated Annealing algorithms, and the Wallacei software uses the NSGA-II algorithm (Deb et al., 2002). Both algorithms produce progressive multi-objective evolutionary processes to achieve a diverse Pareto optimal set within a limited time span and with reasonable computer equipment (Navarro-Mateu et al., 2018).

2.3. Multi-objective optimisation setup

To set up a generative computational model to work with the multi-objective optimisation process, a basic typology of a fabric canopy was constructed within the digital model of the urban site environment. A surface was created in between four coordinate points. The location of these points was set as variable inputs to be tested, constrained to a maximum boundary area on site. An internal ‘funnel’ geometry was created automatically by scaling and extruding the perimeter polyline, and removing the faces oriented towards the open space (Fig. 2a). Using the real-time physics simulation tool Kangaroo (Piker, 2013), a double-curved minimal surface geometry was generated (Fig. 2b). This geometry and material system was chosen as tensile fabric structures can span large areas with minimal use of material. The system also allows easy installation on site, and removal and storage if required.

The experimental setup required the translation of spatial social relationships and environmental performance goals into a set of mathematical variables, relating to the parametric definition and evaluation of our architectural geometry. The open nature of the Rhino/Grasshopper parametric environment allows to construct such complex definitions, as it enables several third-party plugins to be combined, connecting the data flows of generating the phenotypes with tools and rulesets to evaluate their performance.

As part of the parametric definition of the genotype, we set up four coordinate points relative to a chosen site boundary area, shown as x_1,y_1,x_2,y_2,x_3,y_3,x_4,y_4 in Fig. 2. The values were normalised between 0 and 1. With two decimal places set in the definition, this resulted in a total of 808 slider values and 1.1 x 1016 possible variations.

As fitness objectives, four sets of parametric relationships were set up, using a sun analysis tool, surface area calculation and two sets of line-of-sight relationships which were checked for connectivity or blocking. The general aim was to optimise the location and shape of the canopy in relation to three existing benches. Simplified volumes were modelled to represent seated human body positions on the benches, and their surfaces were used to analyse whether any direct sunlight was received. Surfaces were modelled to represent the human eye level zone of seated people on the benches, and line-of-sight relationships were defined with reference to important elements in the surroundings (Fig. 3). The following Fitness Objectives (FO) were defined to optimise the form finding process towards several performance goals for the canopy:

1. **FO1: Maximise shading** for people seated on the three selected benches located across from each other, thus stimulating social interaction by inviting people to sit face to face. Weather data for Hong Kong was imported in the Ladybug plugin, and the analysis date was set to September 15, 11.00 a.m.—15:00 p.m. April and September have the lowest sun vectors and represent the start and end of the period where shading is required for comfortable outdoor seating.
2. **FO2: Minimise the total surface area** of the canopy geometry, to reduce overall project cost in relation to the use of materials, and wind loads which would need to be accommodated for with the steel structure. The definition evaluated the surface area of the final mesh shown in Fig. 2b. A minimum surface area of 15 m^2 was set to avoid too many unusable results.
3. **FO3: Minimise 'being seen'** by blocking views from the shopping gallery and lower rows of residential windows directly behind the site, to improve the feeling of privacy. People will feel less 'overlooked' and the solid canopy behind them provides a feeling of strategic safety, creating a more intimate space. The definition counted the number of viewing lines intersecting with the canopy geometry, shown as blue lines in Fig. 3.
4. **FO4: Maintain 'seeing'** lines of sight to the key public spaces around the benches, to allow for the seated persons to keep an eye on their surroundings, to increase a feeling of safety and control. The definition counted the number of blocked viewing lines towards the main entrances to the public space, shown as red lines in Fig. 3.

A current limitation of the Wallacei plugin is that while it enables to define a larger number of fitness objectives, its algorithm NSGA-II was developed for two objective optimisation problems. As in our study, FO3 and FO4 are set up as constraints, the simulation is expected to produce correct near-optimal results. Future development of our methodology will involve exploring other evolutionary

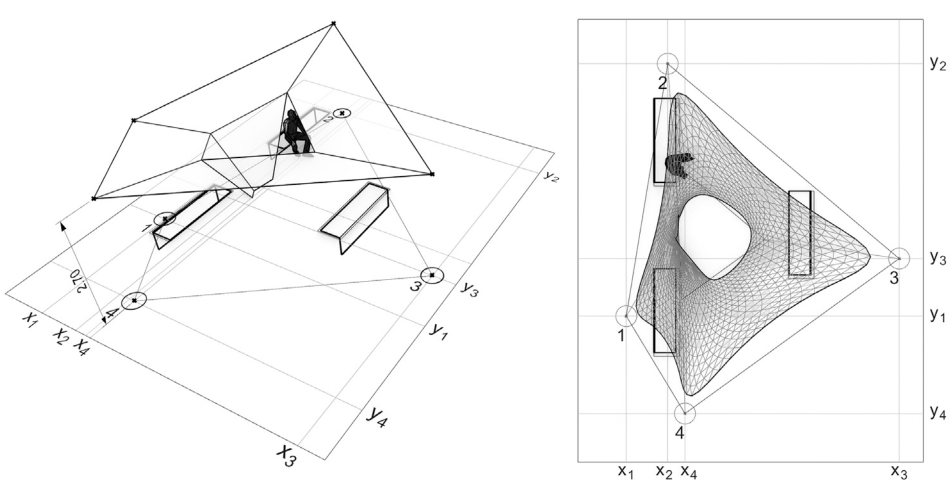


Fig. 2 Generative canopy typology setup, using four points within a predefined boundary area. The double curvature is generated by the physics engine and a smoothing operation is performed on the final geometry.

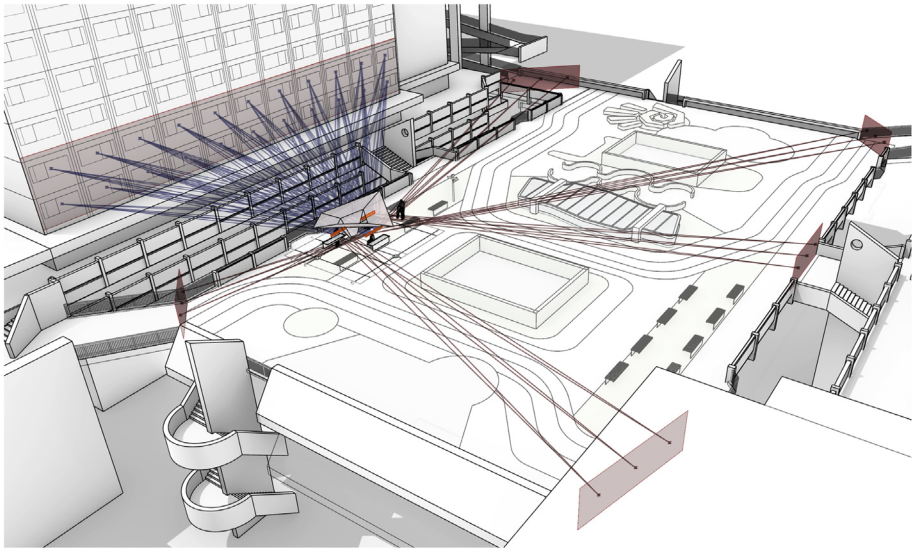


Fig. 3 The canopy basic surface typology at three existing benches, human eye level zones of seated people on the benches (shown in orange), and lines of sight relationships in relation to elements in the surroundings.

algorithms developed for multi-objective optimisation with three or more objectives.

2.4. Evolutionary process results

The evolutionary algorithm in the Wallacei plugin was configured to produce 50 individuals per generation (containing random variations), to be evaluated and optimised across 100 iterative generations. This produced a population size of 5000 individuals within a computing time of around 8.5 min. This fast simulation time allowed for many rounds of iterative testing and improvement of the parametric setup, refining the evaluation mechanisms of the performance indicators, and simulation settings to balance between exploration, and selection and optimisation.

As the multi-objective evolutionary solver tried to optimise all four fitness objectives simultaneously and in parallel, the resulting population set shows a wide range of geometric variations. This variety is also produced by properties of the algorithm that promotes wide exploration of the solution space, to avoid premature convergence towards a local optimum (Navarro-Mateu et al., 2018). Fig. 4 shows a selection of individuals from generation 0, 19, 39, 59, 79 and 99.

The software calculates and visualises the Standard Deviation (SD) values for each generation, plotting blue curves for advanced generations on top of red curves which represent the earlier sets. A solid cluster of blue lines indicate convergence towards an optimised solution space, which occurs in our experiment with fitness objective 1, 2 and 3 (Fig. 5). Fitness objective 4 (maintaining 'seeing' lines towards the surroundings) does not transition to an optimum as this is set up as a 'binary condition', promoting solutions that do not block any viewing lines towards the surroundings, which most of the more advanced solutions can achieve well. The software also plots trendlines for the Standard Deviations and Mean Values (Fig. 5, right side). Gradually increasing SD values and decreasing Mean Values indicate increasing fitness per generation, which in our experiment was achieved gradually for fitness objective 1, while 2 and 3 achieved this more rapidly as the parametric setup made these criteria easier to achieve.

The multi-objective optimisation software stores the performance data of all individuals and ranks them accordingly. The highest-ranking individuals for each of the fitness objectives can be retrieved, giving insights in which specific geometry solutions fulfil those objectives most effectively. Fig. 6 shows the phenotype, performance data and radar charts for each fitness objective: FO1 Shading, FO2: Minimise Surface Area, FO3: Blocking 'being seen', FO4: Maintain 'seeing' lines. Visualisation of the sun hours calculation on the seating area surface produced by Ladybug are also shown. The highest-ranking phenotype for FO1 shows a large canopy surface area covering most of the benches so there is only a small amount sun hours received. The second phenotype, optimised to FO2, does not show the smallest surface area, but an area close to the minimum required surface of 15 m^2, as set up in the algorithm. The third geometry shows optimal blocking of 'being seen' lines (FO3), resulting in a flat and narrow surface parallel to the façade which' window sightlines are aimed to be blocked. The fourth phenotype is a random sample of the solution range, as most individual satisfy the criteria for maintaining surveillance from the benches (FO4).

While the specialised individuals give insights into how separate fitness objectives could be achieved, we are mainly interested in final geometry solutions that strike a balance between trade-offs. As some of the objectives are contradictory, there is no 'perfect' solution that satisfies all criteria. Wallacei offers two methods for identifying solutions that perform well across all objectives. The Relative Difference option aims to find individuals that perform at an equilibrium between objectives, while the Fitness Average method could produce a solution that prioritises one objective by weakening others (Navarro-Mateu et al., 2018). In our process (Fig. 7), the Relative Difference solution was

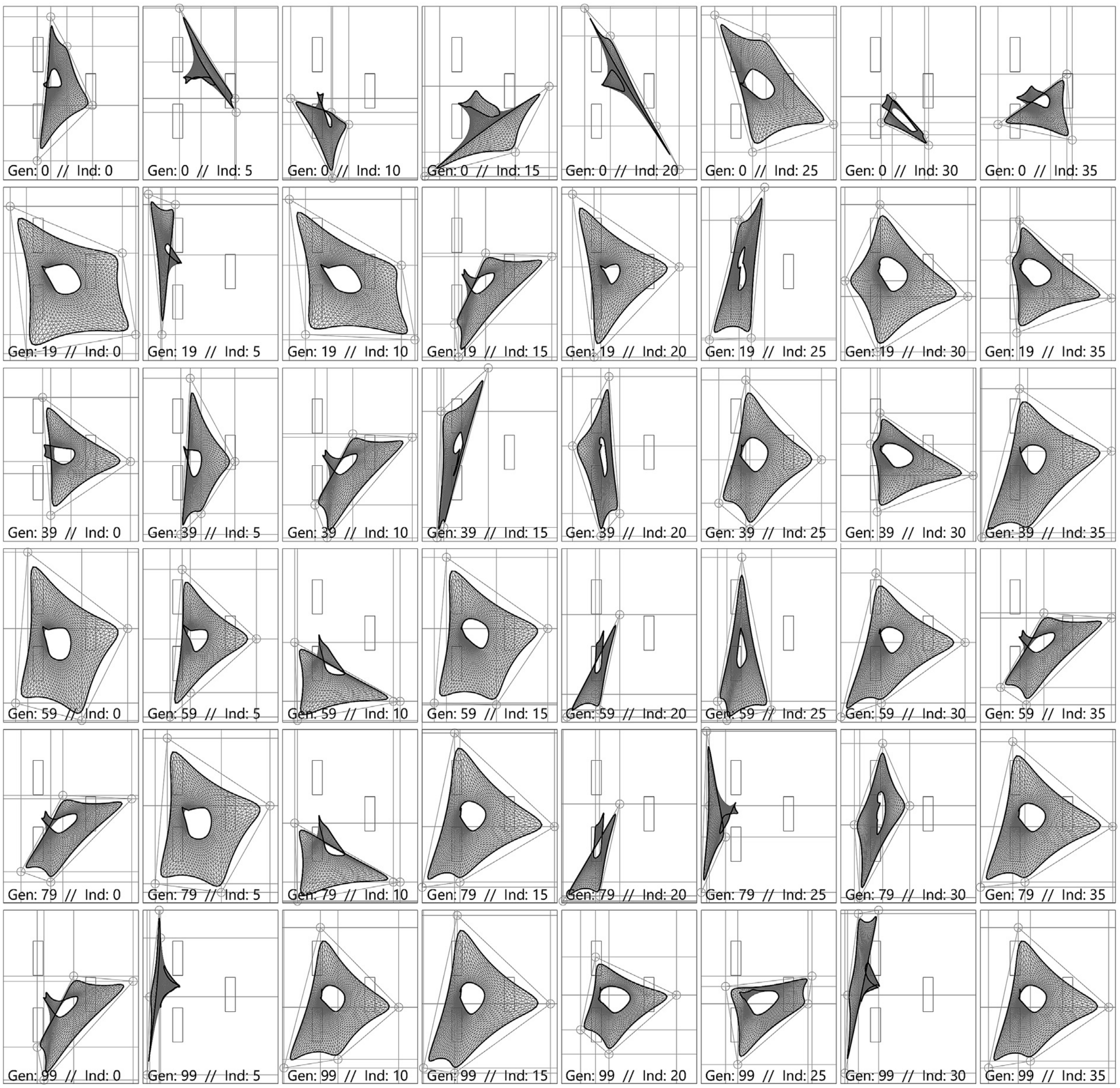


Fig. 4 A selection of individuals from different generations of the multi-objective optimisation process.

less suitable as the sunlight access to the seating was not fully blocked, while the Fitness Average solution performed best in relation to blocking sun, 'being seen' viewing lines and maintaining surveillance of the surroundings. This solution, Generation 90, Individual 25 was selected for further testing and development. Fig. 8 shows a separate simulation with Ladybug on the surface area surrounding the canopy, to visualise the solar radiation impact on the site on several dates and times of the day. The analysis shows that significant parts of the benches would remain shaded throughout the day, and across different months.

3. Membrane design development

For the further development of the project, a multidisciplinary collaboration phase was started between academics and industry professionals specialising in lightweight and tension membrane architecture. Using previous experience with the design development, fabrication and installation, the project underwent several stages of refinement by considering the material, structural and manufacturing complexity of the design.

The first stage of design development was to recreate the surface geometry produced in Rhino/Grasshopper and Kangaroo with the specialised software EASY by Technet, a Germany-based company that supports the development of Membrane- and Cable Net Structures. The canopy design was remodelled by inputting the four control points generated by the multi-objective optimisation process, and the two lower attachment points resulting from the 'funnel' typology encoded in the parametric definition. Fig. 9 shows the digital model of the surface, simulated as a membrane in tension combined with steel edge cables. The software performs its own form finding process, based on dynamic relaxation of the surface in between the

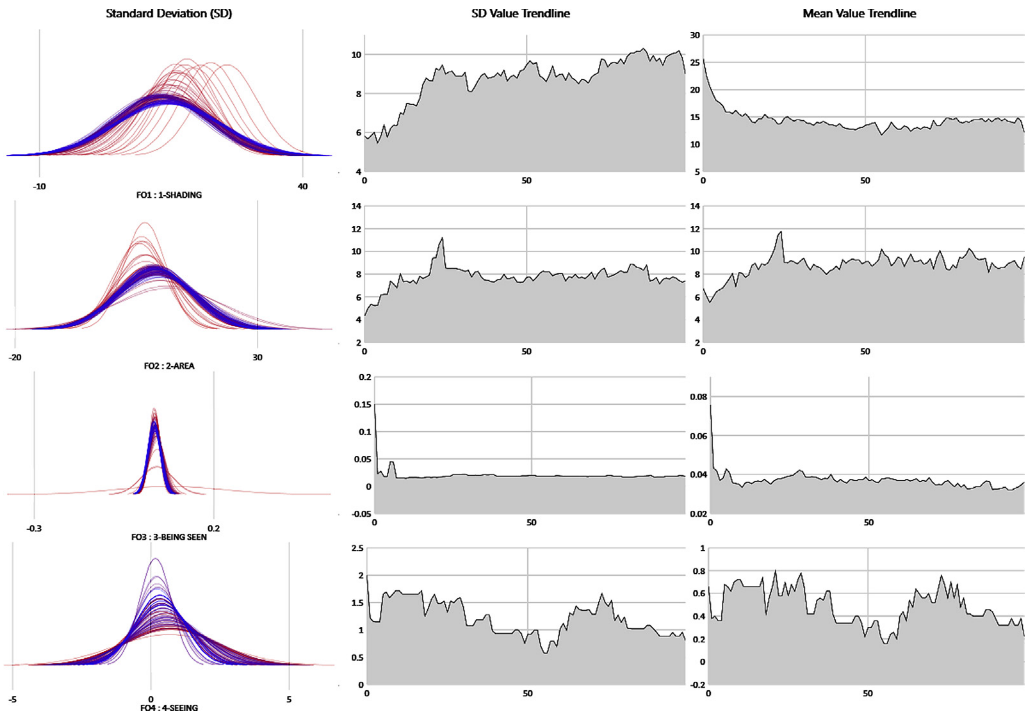


Fig. 5 The Standard Deviation and Mean Values plot lines and trendlines.

attachment points and in relation to pretension forces applied at those points. Higher pretension forces and straighter edge curves would generate large reaction forces which the supporting structure should be able to accommodate. Hence, a balanced solution was found between the final surface curvature and manageable tension forces. The geometries produced by EASY were exported back to Rhino3D and Grasshopper during several stages of the development process, to check whether the required sun shading function was still achieved as the geometry was evolving towards a constructable solution.

The specialised software also allowed for several types of calculations relating to further materialisation. The edge cable forces calculation informed the specification of a steel cable diameter of 10 mm. The calculation of stresses and deformation of the membrane surface informed the selection of a range of suitable fabric types. As the scale of the project was relatively small compared to the range of projects seen in

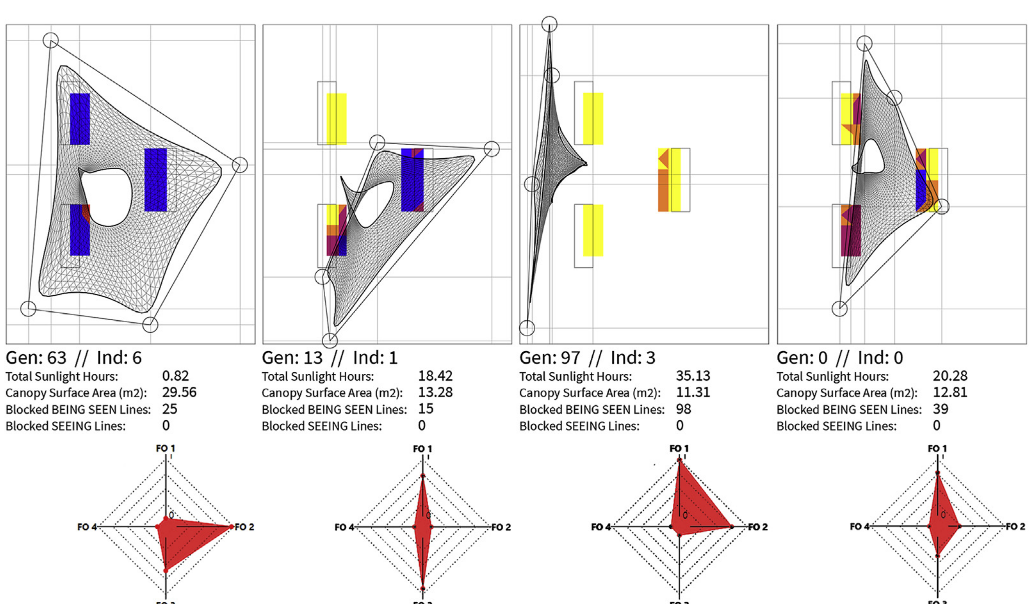


Fig. 6 Top ranked individuals in relation to each of separate fitness objectives.

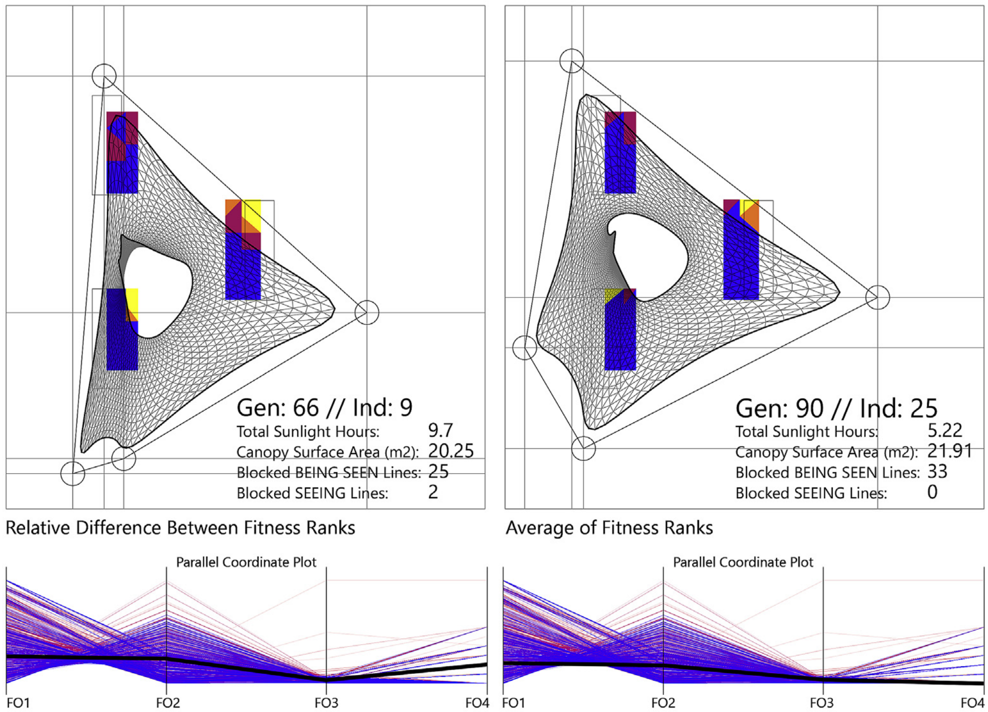


Fig. 7 Highest ranked individuals and the Parallel Coordinate Plot selected by analysing the Relative Difference and Average of fitness ranks.

this field, it was determined that one of the thinner types of material, Pes PVC type 1 specification could be used.

One of the key issues to address during design development was the simulation of wind load scenarios, which would exert forces onto the fabric in addition to the pretension forces. The first aim with tension membrane structures is that wind-related stresses do not greatly exceed the pretension stresses, so that wind gusts do not result in major deformations of the fabric. The resulting stress is kept well within the material properties. The second aspect of considering wind load scenarios is to determine the reaction forces that the supporting structure would need to accommodate. Besides the pretension reaction forces, the EASY software enables the calculation of additional reaction forces based on a given wind speed.

As the project described here classifies as a mobile installation, it does not need to withstand the highest wind loads associated with typhoons under the Hong Kong

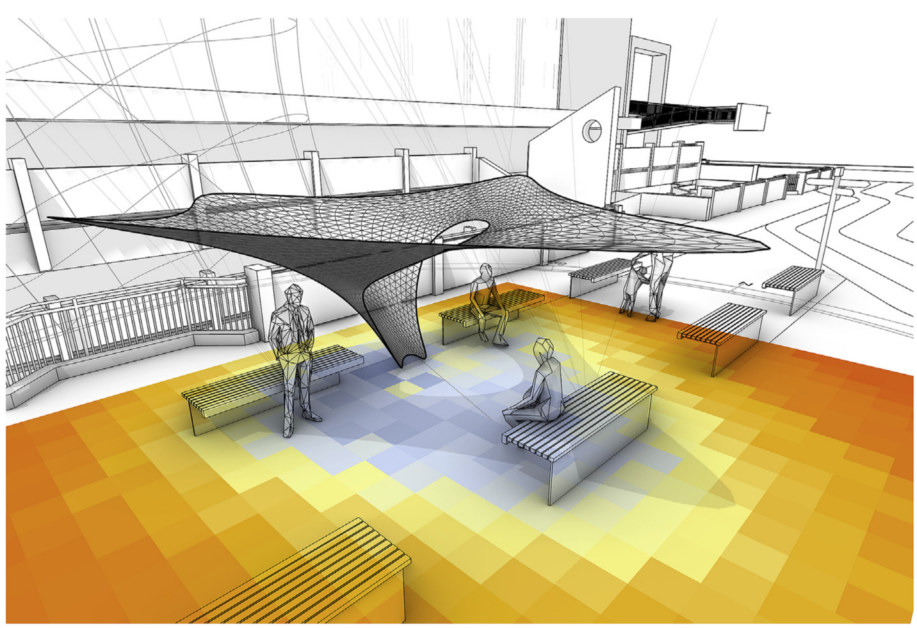


Fig. 8 Testing the selected canopy solution through solar radiation analysis (15 April, 11:00—12:00).

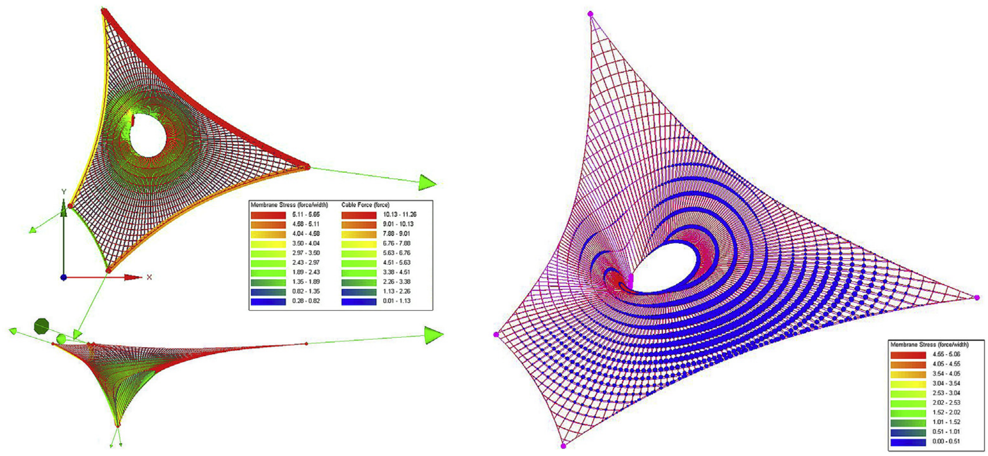


Fig. 9 Form-finding and material behaviour analysis in Technet software for membrane structures.

regulations. These high winds typically occur in the 'typhoon season' which covers August and September. The site managers of the installation agreed that the fabric structure would be taken down during this season, or when strong winds of over 15 m/s (30 knots, 54 km/h) would be forecasted. Analysing weather data from the Hong Kong observatory, we found that average wind speeds between 8.0 and 13.8 m/s occur 0.3% annually, rising to 1.6% when only looking at the months of June/July/August. Average wind speeds higher than 13.8 m/s occur close to 0% during non-typhoon conditions. However, as the Observatory documents sustained wind speeds calculated as 10-min Mean values, the maximum wind gust speeds need to be considered, which are instantaneous and usually higher than the sustained wind speeds. This variation informed the safety factors applied to the design of the steel structure, as well as the material specifications and detailing.

4. Steel structure design

As previously described, some of the requirements and early design decisions for the project included the capability to have a freestanding, movable and demountable structure that does not requires alterations to the site. As support structure for the fabric canopy, a centralised frame principle was chosen, which would also create unobstructed open space around to invite socialising.

4.1. Initial structure design testing

Following from the previous stage of design development, the six anchor points and membrane geometry were defined and considered 'frozen' for the next stage of structure design. The reaction forces were known, as well as the specific vector directions of these forces. Based on studying precedents and previous experiences, it was decided that the central structure would be a truss-like spatial frame, constructed out of Circular Steel Hollow Sections (CHS). In an initial design phase, several different design configurations were developed and tested in conversation with the team member with structural engineering expertise.

The structural layouts were modelled manually in the Rhino3D environment and analysed with the cloud-based structural analysis software 'SkyCiv Structural 3D' (Fig. 10). In relation to the detailing and fabrication of the connections, it was determined that all structural members should be of the same diameter. After calculating the uplift and down-press reaction forces in EASY, these vectors were used as input point loads in the structural analysis of the steel frame. For structural safety analysis, the combined stress (bending + shear + torsion) should not exceed 220 MPa, considering a steel structure with S275 grade steel and full-penetration welded joints. A ULS load factor of 1.4 (for wind load) was applied for the structural analysis, in line with the Code of Practice for the Structural Use of Steel (2011). For this temporary structure, the deflection limit was set at 30 mm at the end of the cantilevered branches, and different steel sections were analysed to find the most lightweight solution.

It was found that with a CHS section of 114.3 mm $\times$ 6.3 mm, most of the structure would withstand the stress resulting from the wind load, except one of the cantilevering members that was not aligned with the reaction vector. The structural design was then iterated in order avoid areas of disproportionate stress, and to reduce the sizes of steel sections and overall weight of the structure if possible.

4.2. Structure design optimisation

In the structure optimisation stage, a computational workflow was again used, similar in nature to the multi-objective optimisation process described earlier in this paper. The general aim was to allow the final composition of the spatial frame structure to be informed by an optimisation algorithm, in response to the reaction forces of the pre-designed canopy and the wind forces in Hong Kong. Therefore, a parametric model was set up in Grasshopper3D, based on several three-dimensional coordinate points located between the edges of the canopy and the boundary of the footprint. These nodes were connected by lines, to form a coherent structural framework in a truss configuration containing internal diagonals for lateral stability.

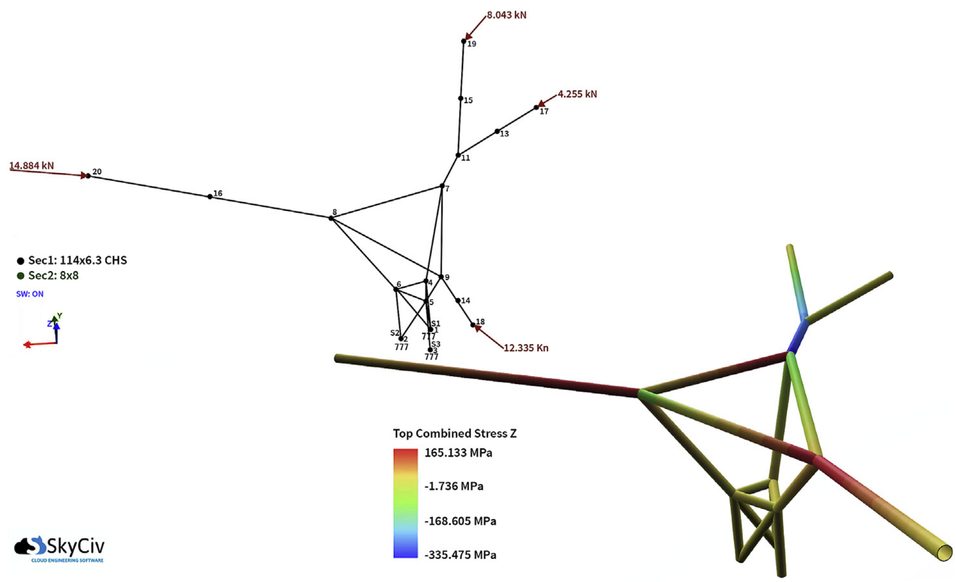


Fig. 10 One of the earlier structural frame options, tested with the fabric tensioning reaction forces.

Within the Rhino/Grasshopper parametric platform, the parametric structural engineering tool Karamba3D (Preisinger, 2013), was used to evaluate the parameterized geometric model, and connect this process to an optimisation algorithm. Similar to the initial design stage, the structure design development required optimisation according to multiple and conflicting fitness objectives. We employed the Octopus plugin developed by Vierlinger (2012), which optimises against multiple goals at the same time according to the Pareto-principle, similar to Wallacei.

The parametric setup used several gene pool components to position the 3D points in space. These gene pools were connected into Octopus to allow the algorithm to run through the spectrum of available solutions (Fig. 11). The objectives of the optimisation were to (1) reduce the overall mass, (2) have a minimal displacement, and (3) have minimal intersections between beams and fabric canopy. The pre-calculated reaction forces as described above were integrated as a load case in Karamba3D, as well as additional load case scenarios simulating horizontal wind directions across the site to optimise the steel structure against all likely load conditions.

In addition to the Evolutionary Structural Optimisation (ESO) process enabled by Karamba3D, the Bidirectional Evolutionary Structural Optimisation (BESO) functionality was used to trace the internal flow of forces through the structure and to evaluate if the least strained elements can be removed. After testing, it was decided to retain all truss elements in order to be able to rely on spatial triangulation principles for lateral stability rather than on the stiffness and welding quality of the joints. Similar to previous projects developed with Karamba3D, it was found that fitness objectives defined towards displacement and self-weight are effective in the optimisation of structural designs through genetic algorithms, and while this produces asymmetrical or irregular layouts, these solutions can be increasingly effective (Preisinger et al., 2011).

The outcomes of the evolutionary process showed structure layouts in which the orientation of the upper beams aligned with the vector orientation of the pre-calculated reaction forces (Fig. 12). From a range of optimised solutions across the Pareto front within Octopus, a solution was picked based on aesthetic considerations and constructability, in particular in relation to the geometric complexity of the joints. Consistent checking and feedback from the structural engineering perspective, as well as final evaluation of the design solution in SkyCiv Structural 3D confirmed that the refined structure is more effective in response to the canopy's reaction forces and wind forces than the earlier, manually designed solutions. The computational optimisation has resulted in the elimination of all bending stress within the members. As the real-world setup might still encounter eccentricities, which could result in the buckling of the cantilever members, a final CHS section of 88.9 mm $\times$ 4 mm was prescribed.

The structural design refinement and finalisation demarcates the last stage of the design development process, in which the overall canopy design, material and structural system have been defined in relation to external and internal requirements. Further steps have been undertaken to develop the material detailing, connections and manufacturing sequencing but as the central focus of this paper is on data-driven design, these aspects will be presented in a separate publication. The completion of the architectural, urban and structural design development through the use of computational tools has resulted in a unique and affordable prototype project that is scheduled for manufacturing and installation on site in the near future (Fig. 13a and b). At this stage, we are able to formulate valuable insights and conclusions regarding the difficulties and opportunities around the computational design approach demonstrated in this project, based on our experiences with multi-

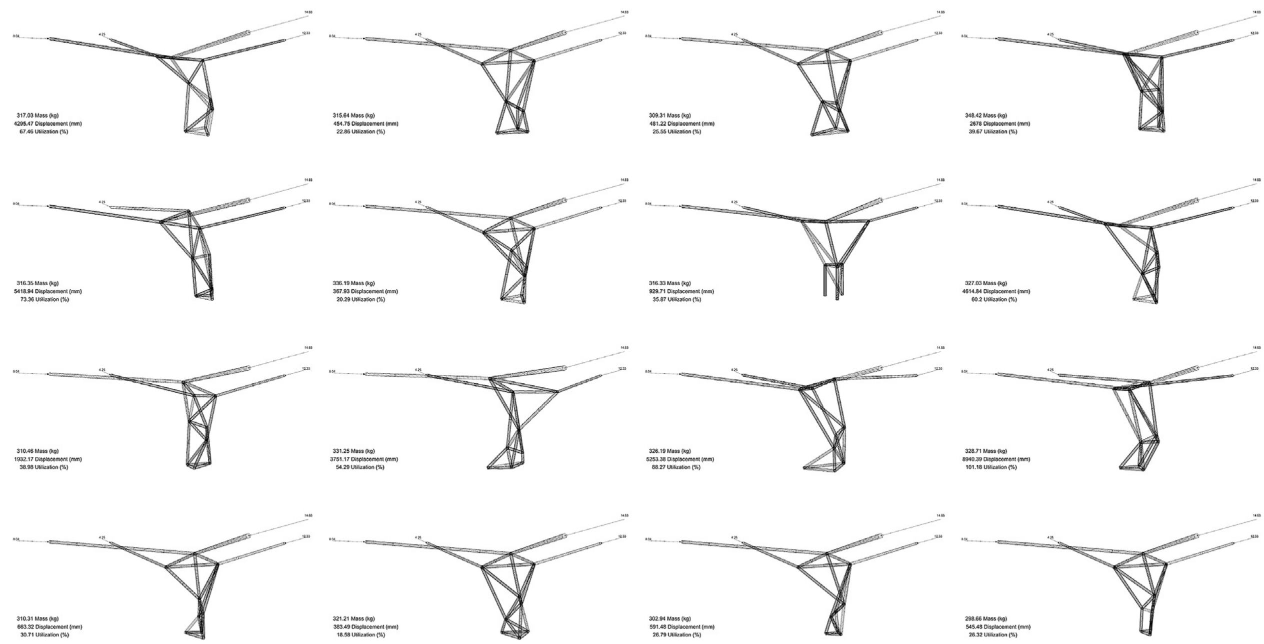


Fig. 11 Some of the layout versions generated during the evolutionary structural optimisation process.

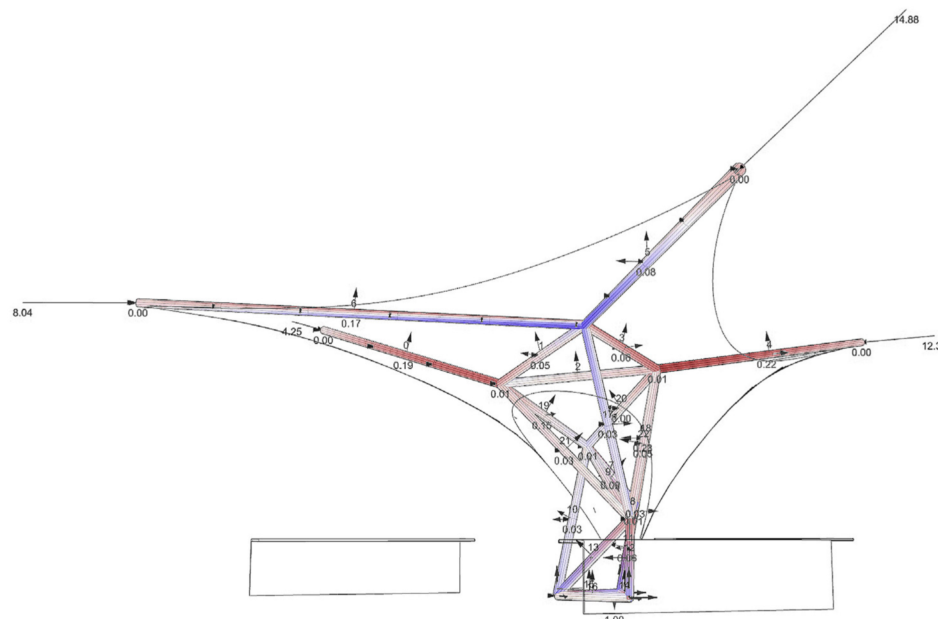
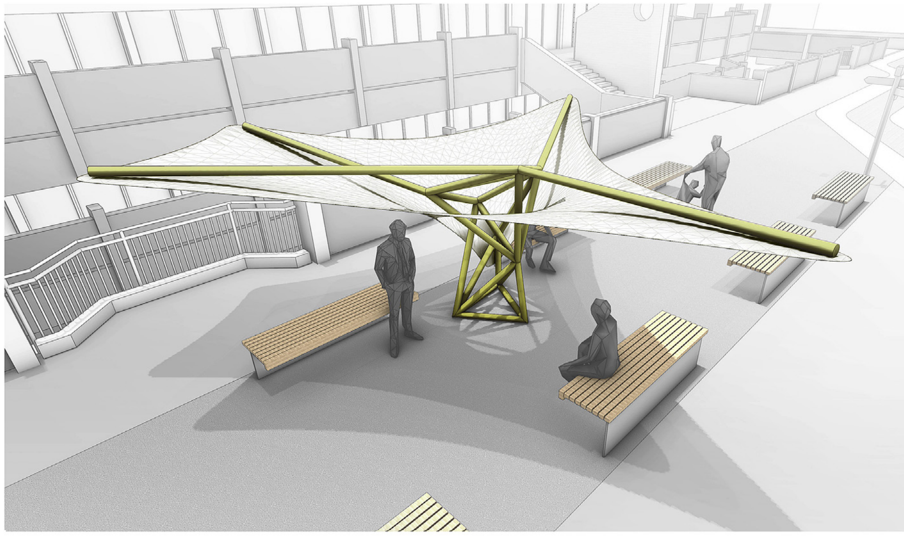


Fig. 12 The final selected configuration, showcasing the alignment of its cantilevering members with the reaction forces.

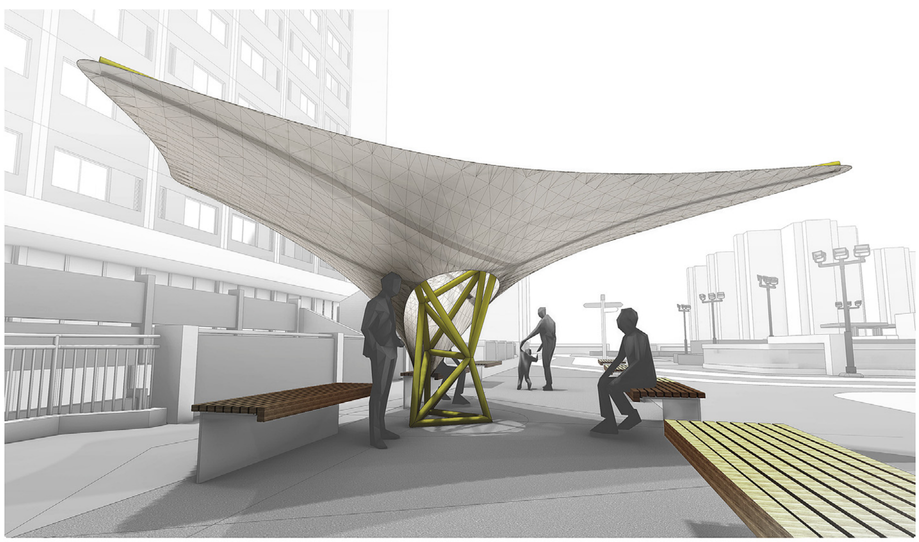
objective optimisation across multiple stages of design development.

5. Discussion

This paper has covered several stages of a project conception and development process that mirror the typical stages in an architectural or urban design project, but that are novel in their integration of computational methods. We have shown how an urban site analysis and a canopy design process can be augmented through digital tools for visibility, environmental and structural analysis, and attempted to demonstrate how additional data can lead to more informed insights about the various performative qualities of architectural structures within their contexts. The existing literature on urban morphology analysis and generative, material performance-based design is separated into discrete domains. There is a wide array of research projects focusing on experimental architecture, yet these projects often focus on novel materials, complex geometry and computational mathematics without quantifying the qualities of the human experiences resulting from these innovative spatial



a



b

Fig. 13 Visualisations of the final design within the site, showcasing the shading qualities and social atmosphere created around the existing benches.

structures. In these demonstrator projects of autonomous digital explorations, explicit relationships to their urban contexts are often lacking. The project we have presented employs a different use of computational tools, by focusing on the spatial and environmental relationships between discrete places and their wider context, and using quantified descriptions of these relationships as design drivers. By focusing on visibility, climate comfort and other placemaking qualities that can promote social interaction, we deploy computation in a human-centric design process, aimed at producing solutions for basic human needs.

The use of evolutionary algorithms and multi-objective optimisation processes present a powerful new toolkit for architectural and urban design problems, incorporating more of the complex performance criteria that play a part in the real-world complexity of human experiences and interactions in public spaces. In our project, the objectives and range of possible outcomes was still quite limited in relation to the possibilities of these new tools, due to the self-imposed limitations of the chosen typology and material system. The main goals of achieving sun shading with a minimum amount of material, while improving local privacy levels and maintaining opportunities to survey the surroundings, might have been solved in a 'manual' digital modelling and testing process. The value of the multi-objective optimisation workflow, however, is not in producing an outcome that works, but in generating all possible solutions that could achieve satisfactory outcomes in different ways. The process is faster, more precise and more transparent than traditional design methods as the solution space is comprehensively explored, analysed and visualised. This helps to understand the way in which design solutions represent a balanced solution, or compromise, between a range of ambitions around a project, which are often contradictory. During the canopy design process, detailed insights in the amount of sunlight or shadow created at the chosen site allowed to make informed decisions about the shape and size of the structure, understanding the trade-off between production cost and placemaking benefits. The use of data-driven methods allowed us to evaluate how (and how many) people would be able to experience improved climate comfort and social interaction opportunities in relation to the required budget for construction. It is easy to see how the open-ended and data-driven nature of the digital workflow explored here can be expanded to encompass more complex architectural or urban design problem, and their evaluation criteria.

6. Conclusions

In this paper, we have offered a reinterpretation of the notion of architectural 'form-finding', from a material performance based concept with a single optimisation criterion to a multi-faceted concept incorporating structural, social and environmental performance. The public seating canopy project discussed here represents an experimental process in enhanced digital design methods using recent computational tools, as part of a larger research agenda around data-driven processes for socially performative public structure and spaces. By making the principles of social interaction and climate comfort in outdoor public spaces explicit, spatial and mathematically quantified, it is possible to describe the performance of architectural structures more distinctly, and optimise them accordingly. Multi-objective optimisation methods enable a more transparent and insightful exploration and discussion of the different possible solutions to a design problem, highlighting which design objectives could be prioritised, while understanding their impact on other objectives. In our public seating canopy project, a balanced solution between fabrication costs and placemaking benefits was found, paving the way for approval and implementation on site. The developed workflow can be employed for a range of future projects at different sites, using 3D data of the surroundings to generate different site-specific solutions at each location. The current method allows to produce other canopy designs with the same material palette and structural principles, while the geometric typology could be expanded to address larger sites, different social activities and more refined environmental performance functionalities.

From a conceptual point of view towards the methodology explored here, our project is part of the wider shift towards data-driven design in many disciplines, using data analytics, machine learning and evolving modes of human-computer interactions to enhance the quality of design processes and their outcomes. It signals a move from traditional linear and reductionist processes led by a single human designer to a collaborative, multi-disciplinary and integrative process. The computational tools employed here refer to 'the biological paradigm', using digital methods to mimic natural processes of mutation, selection and evolution to generate design solutions that are suitable for contexts that are increasingly understood as complex and dynamic environments. Our future work will continue to explore the implications of these methodologies, focusing on digital methods for analysing urban places, aiming to capture their social and environmental complexities and address these through to generative design, producing meaningful architectural and urban design solutions that contribute to the improved well-being of individuals and communities.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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